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Roll No. ...22UMEC1138

BE III (ME)

3

Dyn. of Mach. (M)

THIRD B.E. EXAMINATION - 2024

MECHANICAL ENGINEERING

(V SEMESTER CBCS)

ME-304 A: DYNAMICS OF MACHINES - (M)

Time - Three Hours

Maximum Marks - 80

- Note:- (i) Attempt any **Five** questions.
(ii) All questions carry equal marks.
(iii) Assume suitable data. if not given and mention it clearly.
(iv) Draw neat sketches, wherever necessary.

1. (a) A riveting machine is driven by a motor of 5 KW. The actual time to complete one riveting operation is 1.8 seconds and it absorbs 15 KN.m of energy. The moving parts including the flywheel are equivalent to 180 kg at 350 mm radius. Determine the speed of the flywheel immediately after riveting it is 540 rpm before riveting. Also, find the number of rivets closed per minute. 8
- (b) Explain procedure to construct turning moment diagram of a four stroke IC engine with neat sketches. 8

2. (a) From the first principle, determine expression for displacement, velocity and acceleration of piston in an IC engine. 8
- (b) In a single - slider crank mechanism, the force acting on slider is 4000 N (inward direction) and coefficient of friction between all the links is 0.3. Calculate the driving torque if the pin diameters at joints O, A and B are 80 mm, 80 mm and 40 mm respectively. The dimensions of links are: OA = 300 mm, AB = 1050 mm and $\angle BOA = 60^\circ$. 8
3. (a) Describe the function of a pivoted-cradle balancing machine with the help of a neat sketch. Show that it is possible to make only four test runs to obtain the balance masses in such a machine. 8
- (b) The following data refer to a two-cylinder uncoupled locomotive: Rotating mass per cylinder is 300 kg; reciprocating mass per cylinder is 350 kg; distance between wheels is 1500 mm; distance between cylinder centres is 650 mm; diameter of treads of driving wheels is 180 mm; crank radius is 350 mm; radius of centre of balance mass 650 mm; locomotive speed is 50 km/hr; angle between cylinder cranks is 90° ; dead load on each wheel is 3.5 tonne. Determine the: (i) balancing mass required in the planes of driving

wheels if whole of the revolving and two-third of the reciprocating mass are to be balanced. (ii) swaying couple (iii) variation in the tractive force (iv) maximum and minimum pressure on the rails; (v) maximum speed of locomotive without lifting the wheels from the rails. 8

4. (a) What is an epicyclic and reverted gear train? Explain with suitable sketches. 8
- (b) In a sun and planet gear train, the sun gear wheel having 85 teeth is fixed to the frame. Determine the number of teeth on the planet and annulus wheels if the annulus rotates 160 times and the arm rotates 120 times, both in the same direction. 8
5. (a) A shaft carries four masses in P, Q, R and S which are placed in parallel planes perpendicular to the longitudinal axis. The unbalanced masses at planes Q and R are 3.2 kg and 2.1 kg respectively and both are assumed to be concentrated at a radius of 28 mm while the masses in planes P and S are both at radius of 50 mm. The angle between the planes Q and R is 110° and that between Q and P is 210° , both angles being measured in counter-clockwise direction from the plane Q. The planes containing P and Q are 280 mm apart those containing Q and R are 550 mm. If the shaft is to be completely balanced, determine: (i) masses at the planes P and S; (ii) the distance between the planes R and S; (iii) the angular position of the mass S. 8

- (b) Explain the terms primary balancing and secondary balancing used for balancing of reciprocating masses in reciprocating engine mechanism. 8
6. (a) Show that for a rack and pinion arrangement with 20° pressure angle, the minimum number of teeth required on pinion to avoid interference is 18. 8
- (b) A worm gear set transmits power between two shafts which are at right angle to each other. The gear ratio is 6 and the worm has 4 teeth of normal pitch 20 mm. The pitch diameter of the worm is 50 mm. Calculate the tooth angles of the worm and worm gear and the distance between the shafts. If the efficiency of the drive is 85 percent, the worm being the driver, find the coefficient of friction between the matching tooth surfaces. 8
7. Write short notes (any four). 4X4
- (a) What do you mean by free body diagram? Explain with suitable example.
- (b) Explain clearly the terms static and dynamic balancing.
- (c) Explain law of gearing.
- (d) What is field balancing of rotors? Explain the procedure.
- (e) Explain the procedure adopted for designing the helical gears.

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Roll No. 1125

B.E. III (Mech.)

4 Dynamics of Mach.

THIRD B.E. EXAMINATION - 2019
(MECHANICAL ENGINEERING)

(V SEMESTER)

ME - 304 A: DYNAMICS OF MACHINES (M)

Time - Three Hours

Maximum Marks - 100

Note:- (1) Answer any Five questions..

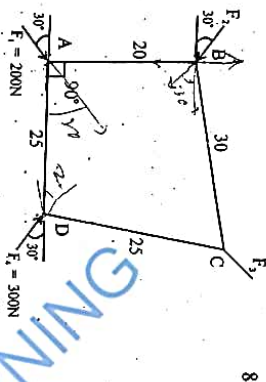
(2) All questions carry equal marks.

(3) Assume suitable data, if not given and mention it clearly.

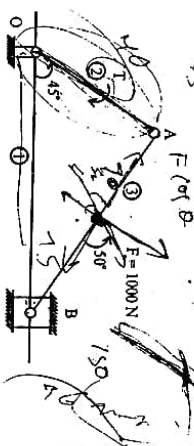
(4) Draw neat sketches, wherever necessary.

3760 1 (Contd.)

1. (a) For the quaternary link shown below determine the magnitude of forces F_1 , F_2 , and direction of F_3 .



(b) Determine the torque T to keep the mechanism shown below, in equilibrium, where, $AC = 70$ mm, $AB = 150$ mm, $CB = 75$ mm, and $OA = 40$ mm.



(2) Derive an expression for crank effort for a given crank position of a slider crank mechanism.

(b) For a horizontal reciprocating engine the mass of reciprocating parts is 120 kg, crank length is 90

3760 2 (Contd.)

mm and engine speed is 600 rpm. The mass of connecting rod is 90 kg, length between centres is 450 mm, the distance between the centre of mass from big end centre is 180 mm and radius of gyration about an axis through centre of mass is 150 mm. Find the magnitude and direction of the inertia torque on the crankshaft when the crank has turned 30° from the inner-dead centre.

3. Explain with help of neat sketch the turning moment diagrams for single cylinder and multi-cylinder internal combustion engine.

(b) A punching machine carries out 6 holes per minute. Each hole of 40 mm diameter in 35 mm thick plate requires 8 Nm of energy/mm² of the sheared area. The punch has a stroke of 95 mm. Find the power of the motor required if the mean speed of the flywheel is 20 m/s. If total fluctuation of speed is not to exceed 3% of the mean speed, determine the mass of flywheel. Also sketch the turning moment diagram for one complete cycle.

(a) With help of neat sketch define and explain the various pitches and pressure angles in helical gears.

3760 3 (Contd.)

- (b) For a right angle spiral drive the normal pitch is 10 mm, the distance between the axes is approximately 150 mm and the speed ratio is 5:2. The frictional angle is 6° and the efficiency is to be maximum.

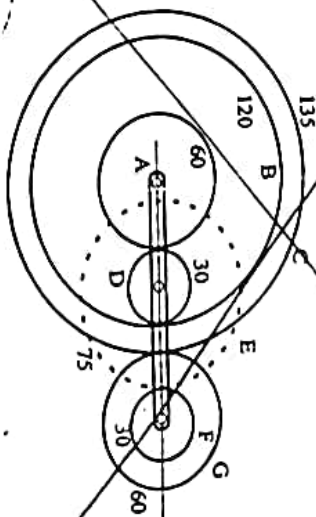
Find

- the apparent circular pitch,
- the spiral angle,
- The number of teeth,
- The exact centre distance and pitch circle diameters of the two wheels and
- the efficiency of the drive.

Explain with help of neat sketch various types of gear trains.

The number of teeth on each gear of the gear teeth are shown in the figure. Gears D-E and F-G are compound gears. D gears with A and B; E gears with F; and G gears with C. If the wheel A is fixed and the arm makes 2 revolutions

(Contd.)



clockwise, find the revolutions of B and C. If the arm is applied a turning moment of 1 kNm, determine the turning moment on the shaft supporting the wheel C.

For an engine prove that the resultant unbalanced force is minimum

When half of the reciprocating masses are balanced by rotating masses, i.e. when $c = 1/2$.

Explain the effects of partial balancing in locomotives.

Write short notes on any four:

- Interference and undercutting
- Balancing of V-engine.
- Law of gearing.
- Dynamically equivalent link.
- Cycloidal profile of teeth.

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5

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4

14

6

12

4x5

8

- (i) The addenda on the wheel and the pinion.
(ii) The path of contact.
(iii) The maximum velocity of sliding at engagement and disengagement of a pair of teeth.
(iv) Contact ratio. (10)
- 6- (a) Determine the unbalanced forces and couples in case of following in-line engines. (10)
(i) Two-cylinder engine.
(ii) Four-cylinder four-stroke engine.
(iii) Six-cylinder four-stroke engine. (10)
(b) A single-cylinder reciprocating engine has a reciprocating mass of 60kg. The crank rotates at 60 rpm and the stroke is 320mm. The mass of the revolving parts at 160 mm radius is 40kg. If two-thirds of the reciprocating parts and the whole of the revolving parts are to be balanced, determine the balance mass required at a radius of 350mm.
(ii) Unbalanced force when the crank has turned 50° from the top-dead centre. (10)
- 7- (i) Write short notes on any four.
(ii) Procedure to construct turning moment diagram of a four-stroke IC engine. (14)
(iii) What do you mean by force balancing of linkages? How is it achieved? Explain. (2)
(iv) State and explain D'Alembert's principle. (2)
(v) Law of gearing. (2)
(vi) What is the principle of virtual work? Explain. (5x4=20)
(vii) Undercutting of gears. (2)

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Roll No.15MEE21522...

B.E. (Mech.)
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Dynamics of Mach.
THIRD B.E. EXAMINATION-2018
(MECHANICAL ENGINEERING)
(V SEMESTER)

ME-304A: DYNAMICS OF MACHINES (M)

Time - Three Hours
Maximum Marks - 100

Note :- (1) Answer any FIVE questions.

- (2) All questions carry equal marks.
(3) Assume suitable data, if not given and mention it clearly.
(4) Draw neat sketches, wherever necessary.

1. (a) The piston diameter of an internal combustion engine is 125mm and the stroke is 220 mm. The connecting rod is 4.5 times the crank length and

has a mass of 50kg. The mass of the reciprocating parts is 30kg. The centre of mass of the connecting rod is 170mm from the crank-pin centre and the radius of gyration about an axis through the centre of mass is 148mm. The engine runs at 320rpm. Determine the magnitude and the direction of the inertia force and the corresponding torque on the crank shaft when the angle turned by the crank is 135° from the inner dead centre. (10)

(b) Describe the graphical method of considering the inertia of the connecting rod of a reciprocating engine. (10)

2- (a) From the first principle, determine expression for displacement, velocity and acceleration of piston in a steam engine. (10)

(b) Determine the required input torque on the crank of a slider-crank mechanism for the static equilibrium when the applied piston load is 1500N. The lengths of the crank and the connecting rod are 40mm and 100mm respectively and the crank has turned through 45° from the inner-dead centre. (10)

3- (a) Prove that the maximum fluctuation of energy

$$\Delta E = E \times 2C_s$$

Where E = Mean Kinetic energy of the flywheel.

C_s = Coefficient of fluctuation of speed. (6)

(b) Discuss the turning moment diagram of a multi-cylinder engine. (4)

(c)

A single cylinder double acting steam engine delivers 185KW at 100 r.p.m. The maximum fluctuation of energy per revolution is 15% of the energy developed per revolution. The speed variation is limited to 1 per cent either way from the mean. The mean diameter is 2.4m. Find the mean and cross-sectional dimension of the flywheel rim when width of rim is twice the thickness. The density of flywheel material is 7200 kg/m³. (10)

4- (a) Explain the difference between simple, compound, and epicyclic gear trains with the help of sketches. What are the special advantages of epicyclic gear trains. (10)

(b) An epicyclic gear consists of a pinion, a wheel of 40 teeth and an annulus with 84 internal teeth concentric with the wheel. The pinion gears with the wheel and the annulus. The arm that carries the axis of the pinion rotates at 100rpm. If the annulus is fixed, find the speed of the wheel; if wheel is fixed, find the speed of the annulus. (10)

5- (a) Find the minimum number of teeth on a pinion of standard addendum and 20° pressure angle to avoid interference between a rack and pinion. (10)

(b) Two 20° full-depth involute spur gears having 30 and 48 teeth in mesh. The pinion rotates at 840rpm. The module is 4mm. Find the sliding velocities at the engagement and at the disengagement of a pair of teeth and the contact ratio. If the interference is just avoided, Determine. (8)

of it is 1.0 m. Also determine the cross-section of flywheel rim. The width to thickness ratio is 1.5 and mass density of flywheel is 7100 kg/m^3

(b) What is meant by static force analysis? Explain with suitable example

In four cylinder In-Line IC engine the mass of reciprocating parts of cylinder 1 and 4 are 100 Kg and that of cylinder 2 and 3 are 173 Kg. If the crank radius is 150 mm, Length of connecting rod is 450 mm and engine speed is 1200 rpm, determine the primary and secondary forces and couples. The cylinders are placed 600 mm apart as shows in fig 3.

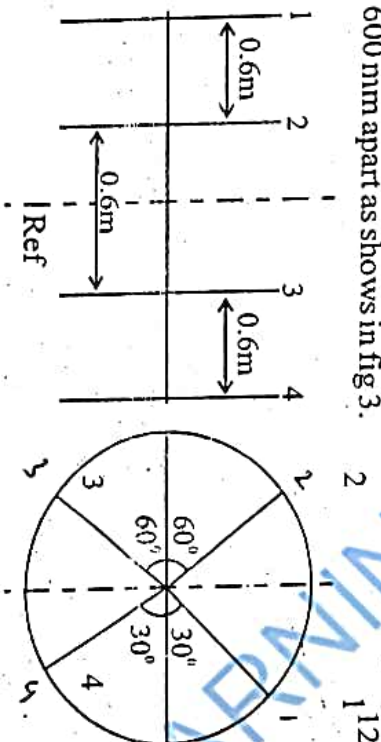


Fig : 3

4x3=12

Write short notes on any **THREE**.

- (i) Direct and Reverse Crank method
- (ii) Pivoted-Cradle Balancing machine
- (iii) Procedure to construct Turning moment diagram of a four stroke IC engine.
- (iv) Dynamically equivalent link
- (v) Law of gearing

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Roll No.

B.E.(Mech.)

4

Dynamics of Mach.

THIRD B.E. EXAMINATION, 2017
(MECHANICAL ENGINEERING)

(V SEMESTER)

ME-304 A: DYNAMICS OF MACHINES (M)

Time - Three Hours

Maximum Marks : 60

Note :- (1) Answer any Five questions

- (2) All questions carry equal marks.
- (3) Assume suitable data, if not given and mention it clearly.
- (4) Draw neat sketches, wherever necessary.

1. Fig-1 shows a mechanism in which a crank AB rotates anticlockwise about A at 70 rpm. The link CD swings about D and connected to the crank pin B by the link BC. The length of other links AB, BC and CD are 0.25 m, 1.0 m and 0.75 m respectively. The connecting link BC has a mass of 14 Kg, its centre of gravity G is at the centre of its length and its radius gyration about a transverse axis through G is 0.3 m. The joint B, C, and D are frictionless and link CD is of negligible mass. For the given position in which AB is at 45° from the

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horizontal, find the force exerted at the joints B and C resulting from the inertia of the BC.

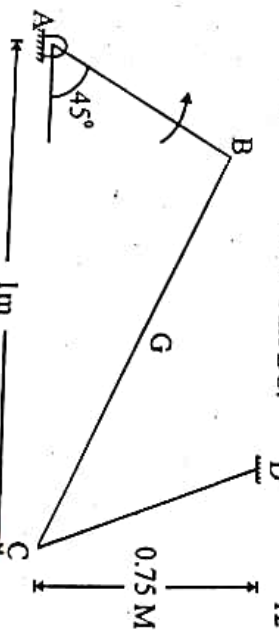


Fig-1

(a) In a slider crank mechanism, the length of stroke is 120 mm and connecting rod is 250 mm long, when the crank rotates at 2000 rpm and travelled 60° from dead centre. Find

- Displacement of piston
- Velocity and acceleration of the piston
- Angular velocity and acceleration of connecting rod.

(b) From the first principle, determine expressions for displacement, velocity and acceleration of piston in an IC Engine.

(c) Explain the phenomenon of interference as experienced in involute gears. State the conditions under which the interference can be avoided.

(b) A pinion having 30 teeth drives a spur gear having 80 teeth, the profile of the gear is involute with 20° pressure angle, 12 mm module and addendum 10 mm. Find the:

- Length of path of contact
- Arc of contact

(iii) the contact ratio

Also shows that there is no interference between the teeth.

(a) Derive an expression for centre distance between a pair of spiral of gear in mesh.

(b) Fig-2 shows an epicyclic gear train in which gear A drives the internal gear D through compound gear B and C. The number of teeth on gear A is 20 and centre distance between the centres of gears A and B is 300 mm. If the module of all gears is 10 mm and gear C has 80 teeth. Find the speed of the gear D. The arm rotates at 600 rpm in counter clockwise direction and gear A is fixed.

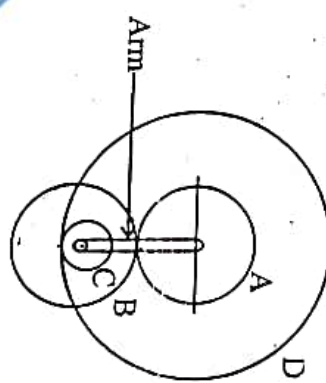


Fig-2

(a) A single cylinder four stroke petrol engine develops 20 kW at 600 rpm. The mechanical efficiency of the engine is 80 percent and the work done by the gases during the expansion stroke is three times the work consumed in compression stroke. If the total fluctuation of speed is not to exceed ± 1.5 percent of the mean speed and turning moment diagram during expansion stroke is assumed to be triangle. Determine the mass of the flywheel if the diameter

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Roll No. 12ME3042

B.E. (MECH.)

Dynamics of Mach.

Third B.E. Examination, 2015
(Mechanical Engineering)

(V-Semester)

ME-304 A : DYNAMICS OF MACHINES (M)

Time—Three Hours

Maximum Marks—60

Note :—(1) Answer any FIVE questions.

(2) Marks allotted to each part of the question are indicated on the right side.

(a) What is law of gearing? Prove it for two mating gears.

(b) The following data refers to a pair of spur gear in mesh having 20° pressure angle involute profile teeth:

Number of teeth on pinion = 20

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Number of teeth on gear = 40
Speed of pinion = 300 rpm

Module = 5 mm

If the addendum on each gear is such that the path of approach and recess are half of their maximum value, find:

- Addendum on gear and pinion
- Length of arc of contact
- Contact ratio, and

(iv) Maximum velocity of sliding of gears. 8

(a) From first principle, derive an expression for minimum number of teeth on gear to avoid interference with pinion. 5

(b) Two shafts A and B are co-axial as shown in Fig. 1. The gear C having 50 teeth is rigidly mounted on shaft A. A compound gear D-E mesh with gear C and internal gear G. Gear D has 30 teeth and meshes with C and gear E having 35 teeth gears

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(Contd.)

with internal gear G. Gear G is fixed and is concentric with shaft axis. The compound gear D-E is mounted on pin which projects from an arm keyed to the shaft B.

Determine:

- The number of teeth on gear G, if module of all gears are same.
- If shaft A rotates at 110 rpm, determine the speed of shaft B.

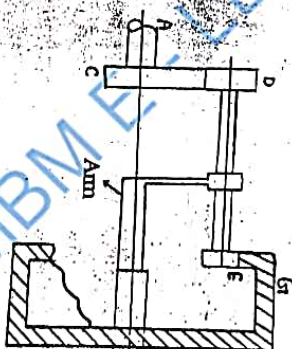


Fig. 1

3. In a four bar mechanism shown in Fig-2 a rotating crank AB driving an oscillating link CD through the coupler link BC. The mass of link BC is 0.8 kg and its radius of

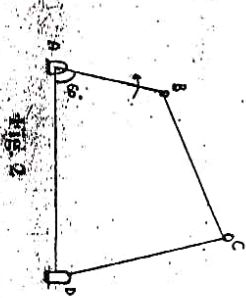
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gyration about an axis through point G is 0.25 m. The distance CG is half of the length of coupler link BC. When crank AB rotates in anticlockwise direction at a constant speed of 500 rpm,

determine :

- (i) Inertia force on link BC and its point of application.
- (ii) Inertia couple at the crank AB.



AB = 300 mm
BC = 400 mm
CD = 500 mm
AD = 600 mm
 $\angle DAB = 60^\circ$

Fig. 2

12

4. (a) Derive an expression of crank effort for a given crank position of a slider crank mechanism. 4

- (b) A punching machine operates at the rate of 600 holes/hr. It does 45 Nm of work per sq. mm of shear, cutting 25 mm diameter hole in 3 mm thick plate. The machine is operated by a constant torque

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motor. The speed of machine fluctuates between 250 and 230 rpm. The frictional losses are 20 % of the work done during punching. The actual punching time per hole is 2 seconds. Sketch the turning moment diagram for one complete cycle. Determine :

- (i) Motor power
- (ii) Maximum fluctuation of energy

- (iii) Mass of the flywheel, if radius of gyration is 500 mm. 8

5. In a slider crank mechanism, the length of stroke is 100 mm and connecting rod is 200 mm long. When the crank rotates at 1000 rpm and has travelled 60° from the dead centre, find :

- (i) Displacement, velocity and acceleration of piston.
- (ii) Derive an expression for displacement, velocity and acceleration of piston. 12

6. Determine the masses to be added at planes X and Y at a radius of 500 mm if the system shown in Fig-3 is to be dynamically balanced. The unbalance masses and their

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eccentricities are given in the following table :

Plane	Mass (kg)	Eccentricity (mm)
A	300	0.2
B	400	0.15
C	450	0.25
D	500	0.3
		12



Fig. 3

7. Write short notes on any TWO.

- (i) Dynamic equivalent link
- (ii) Cycloidal profile of teeth
- (iii) Balancing machine
- (iv) Direct and reverse crank method. $6 \times 2 = 12$

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6

300

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Roll No. 210MECA1154...

BE III (M) .
3

DOM.

THIRD B.E. EXAMINATION, 2023
(MECHANICAL ENGINEERING)
(V SEMESTER CBCS)
ME – 304 A: DYNAMICS OF MACHINES

Time – Three Hours

Maximum Marks – 80

- Note:
- (i) Attempt any Five questions.
 - (ii) Answer should be to the point.
 - (iii) All questions carry equal marks.
 - (iv) Assume suitable data, if not given and mention it clearly
 - (v) Draw neat sketches, wherever necessary.

- 1 (a) In a single-acting four-stroke engine, the work done by the gases during the expansion stroke is two times the work done during compression stroke. The work done during the suction and exhaust stroke is negligible. The engine develops 24 kW at 350 rpm. The fluctuation of speed is limited to $\pm 2.5\%$ of the mean speed on either stroke. The turning-moment diagram during the compression and the expansion stroke may be triangular in shape. Determine the inertia of the flywheel. 8
- (b) Derive the expression for the angular acceleration of the connecting rod of a reciprocating engine. 8
- 2 (a) Draw and explain Klien's construction to determine the velocity and acceleration of the piston in a slider crank mechanism. 8
- (b) The piston diameter of an internal combustion engine is 125 mm and the stroke is 220 mm. The connecting rod is 4.5 times the crank length and has a mass of 50 kg. The mass of

kg respectively. Determine the mass A and the relative angular positions of the four masses if the planes are spaced 500 mm apart. 8

(b) Determine the unbalanced forces and couples in case of following in-line engine:

(i) two-cylinder engine.

(ii) four-cylinder four-stroke engine.

(iii) six-cylinder four-stroke engine. 8

6 (a) Explain the term static balancing and dynamic balancing. 8

State the necessary conditions to achieve them. 8

(b) A three cylinder radial engine driven by a common crank has the cylinders at 120° . The stroke is 125 mm, length of the connecting rod 225 mm and the mass of the reciprocating parts per cylinder 2 kg. Calculate the primary and secondary forces at crank shaft speed of 1200 rpm. 8

Write short notes (any four). $4 \times 4 = 16$

a. Discuss the balancing of V-engines.

b. Explain the principle of virtual work.

c. Explain law of gearing

d. Procedure to construct turning moment diagram of a four stroke engine.

e. Explain the procedure adopted for designing the spur gears.

the reciprocating parts is 30 kg. The centre of mass of the connecting rod is 170 mm from the crank pin centre and the radius of gyration about an axis through the centre of mass is 148 mm. the engine runs at 320 rpm. Find the magnitude and the direction of the inertia force and the corresponding torque on the crankshaft when the angle turned by the crank is 140° from the inner dead centre. 8

- 3 (a) Derive an expression for the velocity of sliding between a pair of involute teeth. State the advantages of involute profile as a gear tooth profile. 8
- (b) A drive is made up of two spiral gear wheels of the same hand, same diameter and of normal pitch of 14 mm. the centre distance between the axes of the shafts is approximately 150 mm. The speed ratio is 1.6 and the angle between the shafts is 75° . Assuming a friction angle of 6° , determine the following: (i) spiral angle of each wheel (ii) number of teeth on each wheel (iii) efficiency of the drive (iv) maximum efficiency. 8
- 4 (a) What is the difference between a simple gear train and a compound gear train? Explain with the help of sketches. 8
- (b) An epicyclic gear consists of a pinion, a wheel of 40 teeth and an annulus with 84 internal teeth concentric with the wheel. The pinion gears with the wheel and the annulus. The arm that carries the axis of the pinion rotates at 100 rpm (clockwise).

Case 1. If the annulus is fixed, find magnitude and direction of the speed of wheel.

Case 2. If wheel is fixed, find magnitude and direction of the speed of annulus. 8

- 5 (a) Four masses A, B, C and D carried by a rotating shaft at radii 80 mm, 100 mm, 160 mm and 120 mm respectively are completely balanced. Masses B, C and D are 8 kg, 4 kg and 7